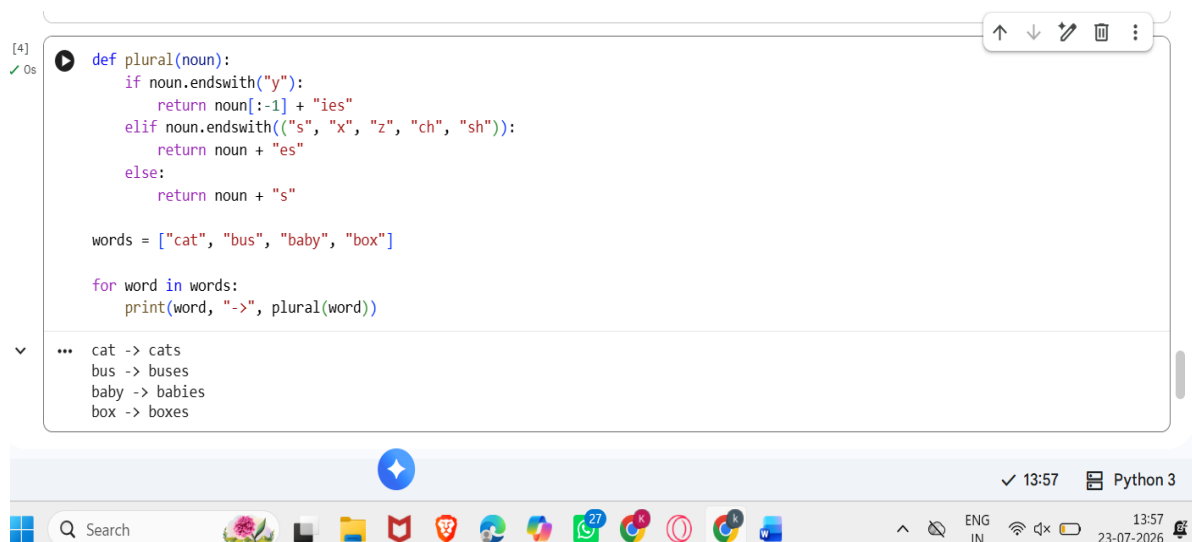


Experiment 4: Finite-State Machine for Plural Forms

Program

```
def plural(noun):  
  
    if noun.endswith("y"):  
  
        return noun[:-1] + "ies"  
  
    elif noun.endswith(("s", "x", "z", "ch", "sh")):  
  
        return noun + "es"  
  
    else: return noun + "s"  
  
words = ["cat", "bus", "baby", "box"]  
  
for word in words:  
  
    print(word, "->", plural(word))
```

Output



```
[4]  
✓ 0s  
def plural(noun):  
    if noun.endswith("y"):  
        return noun[:-1] + "ies"  
    elif noun.endswith(("s", "x", "z", "ch", "sh")):  
        return noun + "es"  
    else:  
        return noun + "s"  
  
words = ["cat", "bus", "baby", "box"]  
  
for word in words:  
    print(word, "->", plural(word))  
  
... cat -> cats  
    bus -> buses  
    baby -> babies  
    box -> boxes
```

13:57 Python 3
23-07-2026